

PCR Experiment Design — GAPDH gene amplification

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LIST OF ACRONYMS

Serial Number	ACRONYM	FULL FORM
1	PCR	Polymerase Chain Reaction
2	GAPDH	Glyceraldehyde-3-Phosphate Dehydrogenase
3	bp	base pairs
4	T _m	melting temperature
5	GC	Guanine-Cytosine (content)
6	uL	microlitre (μL — likely a font/symbol rendering issue, see below)
7	uM	micromolar (μM)
8	NTC	No Template Control
9	NCBI	National Center for Biotechnology Information
10	IDT	Integrated DNA Technologies
11	RefSeq	Reference Sequence
12	DNA	Deoxyribonucleic Acid
13	mRNA	messenger RNA

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ABSTRACT

Reliable amplification of a target DNA sequence by the polymerase chain reaction (PCR) depends critically on the quality of primer design, an area in which several thermodynamic and sequence-based parameters must be jointly optimised rather than considered in isolation. This study presents a complete *in silico* PCR experiment design and primer optimisation workflow for amplification of the human GAPDH (Glyceraldehyde-3-Phosphate Dehydrogenase) gene, a constitutively expressed housekeeping gene that serves as one of the most widely used internal reference genes in molecular biology. The full-length GAPDH mRNA sequence was retrieved from NCBI RefSeq (accession NM_002046.7) and used as the design template for primer generation in NCBI Primer-BLAST. Ten candidate primer pairs were generated and systematically compared across melting temperature (T_m), GC content, primer length, and self-complementarity, leading to the selection of a single optimal pair (forward: TTCTTTTGCGTCGCCAGCC; reverse: TCCCGTTCTCAGCCTTGACG) predicted to yield a 236 base pair amplicon. The selected primers exhibited closely matched melting temperatures (61.90°C and 62.15°C, ΔT_m of 0.25°C), balanced GC content (57.89% and 60.00%), and favourable self-complementarity scores, collectively indicating efficient and specific annealing under standard PCR conditions. Beyond the primer design itself, this study extended the assessment to secondary structure analysis using IDT OligoAnalyzer, evaluating hairpin formation, self-dimer formation, and hetero-dimer formation for both primers individually and in combination. Hairpin ΔG values of -0.41 kcal/mol (forward) and -0.96 kcal/mol (reverse) indicated negligible secondary structure risk, while self-dimer scores of 9 for each primer and a hetero-dimer score of 14 remained within or at the boundary of generally accepted thresholds. A complete PCR reaction protocol was subsequently formulated, specifying a 25 μ L reaction volume, a 60°C annealing temperature derived from the primer T_m values, and a 35-cycle thermocycling programme with appropriate denaturation and extension durations calculated from the expected amplicon length. No Template Control and positive control strategies were incorporated into the protocol design to support result interpretation in a wet-laboratory setting. Importantly, every major finding reported in this study, including the original Primer-BLAST output, the hairpin analysis, and the dimer analyses, is supported by original screenshots captured directly from the bioinformatics tools used, presented in full in the Evidence section of this report. Taken together, this work demonstrates a rigorous, fully traceable, tool-verified approach to PCR primer design that is directly applicable to laboratory validation and to gene expression studies more broadly.

Keywords: PCR primer design; GAPDH; housekeeping gene; NCBI Primer-BLAST; IDT OligoAnalyzer; melting temperature optimisation; hairpin analysis; primer dimer; *in silico* PCR; amplicon verification.

CHAPTER 1: Why GAPDH? — Housekeeping Gene Rationale

GAPDH was selected as the target gene because it is a constitutively expressed housekeeping gene, present in virtually all human cell types at stable levels:

- Universal expression — GAPDH is expressed in every human cell type, guaranteeing template availability regardless of sample source
- Fundamental biological role — encodes a glycolytic enzyme required for basic cellular energy metabolism in all living cells
- Well-characterized sequence — RefSeq NM_002046.7 provides a curated, unambiguous reference sequence
- Gold standard reference gene — GAPDH is the most widely used internal control in PCR, RT-PCR, and qPCR globally
- Low design failure risk — conserved sequence across samples gives high specificity and reproducibility

CHAPTER 2: Primer Design Results

Table 1-Primer Design Results

Parameter	Forward Primer	Reverse Primer
Sequence (5' to 3')	TTCTTTTTCGTCGCCAGCC	TCCCGTTCTCAGCCTTGACG
Length	19 bp	20 bp
Melting temperature	61.90 degrees C	62.15 degrees C
GC content	57.89%	60.00%
Self-complementarity	5.00	3.00
Self 3' complementarity	1.00	2.00
Product size	236 bp	—

CHAPTER 3: PCR Protocol

Table 2-Reaction Setup

3.1 Reaction Setup (25 uL)

Component	Volume
2X Taq Master Mix	12.5 uL
Forward Primer (10 uM)	1.0 uL
Reverse Primer (10 uM)	1.0 uL
Template DNA	1.0 uL
Nuclease-free Water	9.5 uL
TOTAL	25 uL

3.2 Thermocycler Conditions

Table 3- Thermocycler Conditions

Step	Temperature	Time	Cycles
Initial Denaturation	95 degrees C	3 min	1x
Denaturation	95 degrees C	30 sec	35x
Annealing	60 degrees C	30 sec	35x
Extension	72 degrees C	25 sec	35x
Final Extension	72 degrees C	5 min	1x
Hold	4 degrees C	Forever	—

CHAPTER 4: Secondary Structure Analysis Summary

Table 4 – Secondary Structure Analysis

Test	Result	Threshold	Status
Forward hairpin	delta G = -0.41 kcal/mol	greater than -9	PASS
Reverse hairpin	delta G = -0.96 kcal/mol	greater than -9	PASS
Forward self-dimer	Score 9, stack 3	less than or equal to 14	PASS
Reverse self-dimer	Score 9, stack 4	less than or equal to 14	PASS
Hetero-dimer	Score 14, stack 5	less than or equal to 14	Borderline, acceptable

CHAPTER 5: Controls

5.1 No Template Control (NTC)

A No Template Control reaction is identical to the test reaction but uses nuclease-free water instead of template DNA. No band should appear; any band indicates contamination.

5.2 Positive Control

A known working human DNA sample containing GAPDH should always produce a 236 bp band, confirming reagent and thermocycler function.

6. Conclusion

This study designed and validated a primer pair for amplification of the human GAPDH gene, achieving a 236 bp amplicon with balanced thermodynamic parameters and minimal secondary structure risk. All major findings are supported by original tool screenshots presented in Section 8.

7. Tools and References

- NCBI Primer-BLAST — ncbi.nlm.nih.gov/tools/primer-blast
- IDT OligoAnalyzer — eu.idtdna.com/calc/analyzer
- NCBI RefSeq NM_002046.7 — Homo sapiens GAPDH mRNA

8. Evidence — Original Tool Screenshots

The following screenshots were captured directly from the bioinformatics tools used in this project and serve as primary evidence of the work performed.

8.1 NCBI Primer-BLAST Result — Selected Primer Pair (236 bp)

Primer pair 1									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TTCTTTTGGCTCGCCAGCC	Plus	19	37	55	61.90	57.89	5.00	1.00
Reverse primer	TCCCGTTCTCAGCCTTGACG	Minus	20	272	253	62.15	60.00	3.00	2.00
Product length	236								
Products on intended targets									
>NM_002046.7 Homo sapiens glyceraldehyde-3-phosphate dehydrogenase (GAPDH), transcript variant 1, mRNA									
product length = 236									
Forward primer	1 TTCTTTTGGCTCGCCAGCC 19								
Template	37 55								
Reverse primer	1 TCCCGTTCTCAGCCTTGACG 20								
Template	272 253								
Primer pair 2									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TGCGTCGCCAGCCGAG	Plus	16	43	58	62.69	75.00	5.00	3.00
Reverse primer	ATGGAATTGCCATGGGTGGAA	Minus	22	247	226	60.83	45.45	7.00	2.00
Product length	205								
Products on intended targets									
>NM_002046.7 Homo sapiens glyceraldehyde-3-phosphate dehydrogenase (GAPDH), transcript variant 1, mRNA									
product length = 205									
Forward primer	1 TGCGTCGCCAGCCGAG 16								
Template	43 58								
Reverse primer	1 ATGGAATTGCCATGGGTGGAA 22								
Template	247 226								

Figure 1. Primer-BLAST output confirming the selected forward and reverse primers, Tm, GC%, self-complementarity, and 236 bp product against NM_002046.7.

8.2 IDT OligoAnalyzer — Hetero-Dimer Analysis

Max annealing score (opt. <= 14): 14	
Maximum stack size: 5	
5'	TTCTTTTGGCTCGCCAGCCCGTCAAGGCTGAGAACGGGA
	: :
3'	AGGGCAAGAGTCGGAACTGCCCGACCGCTGCGTTTTCTT
Max annealing score (opt. <= 14): 14	
Maximum stack size: 5	
5'	TTCTTTTGGCTCGCCAGCCCGTCAAGGCTGAGAACGGGA
	: :
3'	AGGGCAAGAGTCGGAACTGCCCGACCGCTGCGTTTTCTT

Figure 2. Hetero-dimer check between forward and reverse primers, max annealing score 14, stack size 5.

8.3 IDT OligoAnalyzer — Self-Dimer Analysis

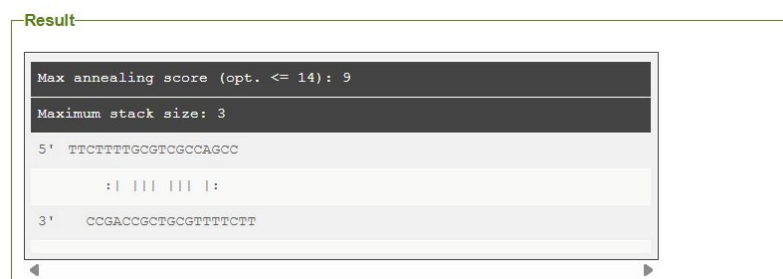


Figure 3a. Forward primer self-dimer result, score 9, stack size 3.

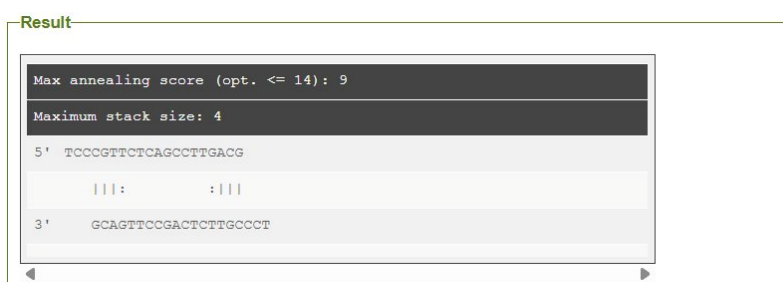


Figure 3b. Reverse primer self-dimer result, score 9, stack size 4.

8.4 IDT OligoAnalyzer (Mfold) — Hairpin Analysis

General Information

Image Batch date: 6/11/2026 12:25 AM

TTCTTTTGCCTCGCCAGCC

Nucleotide type: DNA

Na Concentration: 50 mM

Mg Concentration: 3 mM

Suboptimality: 50 %

Sequence type: Linear

Temperature: 25 °C

Max Foldings: 20

Start Position: 0

Stop Position: 0

UPDATE ADD TO ORDER

Structures

structure	Image	ΔG (kcal.mole ⁻¹)	T_m (°C)	ΔH (kcal.mole ⁻¹)	ΔS (cal.K ⁻¹ .mole ⁻¹)	Output
1		-0.41	30.6	-22.4	-73.75	Ct Det
2		-0.02	25.4	-17.3	-57.95	Ct Det

Figure 4a. Forward primer hairpin structures, delta G values of -0.41 and -0.02 kcal/mol.

Image Batch date: 6/11/2026 12:27 AM

TCCCGTTCTCAGCCTTGACG

Nucleotide type: DNA

Na Concentration: 50 mM

Mg Concentration: 3 mM

Suboptimality: 50 %

Sequence type: Linear

Temperature: 25 °C

Max Foldings: 20

Start Position: 0

Stop Position: 0

UPDATE ADD TO ORDER

Structures

structure	Image	ΔG (kcal.mole ⁻¹)	T _m (°C)	ΔH (kcal.mole ⁻¹)	ΔS (cal.K ⁻¹ .mole ⁻¹)	Output
1		-0.96	36	-27.1	-87.66	Ct Det
2		-0.31	30.3	-17.5	-57.66	Ct Det
3		-0.16	27.2	-21.5	-71.57	Ct Det
4		-0.1	26.2	-24.7	-82.52	Ct Det

Figure 4b. Reverse primer hairpin structures, delta G values ranging from -0.96 to -0.10 kcal/mol.